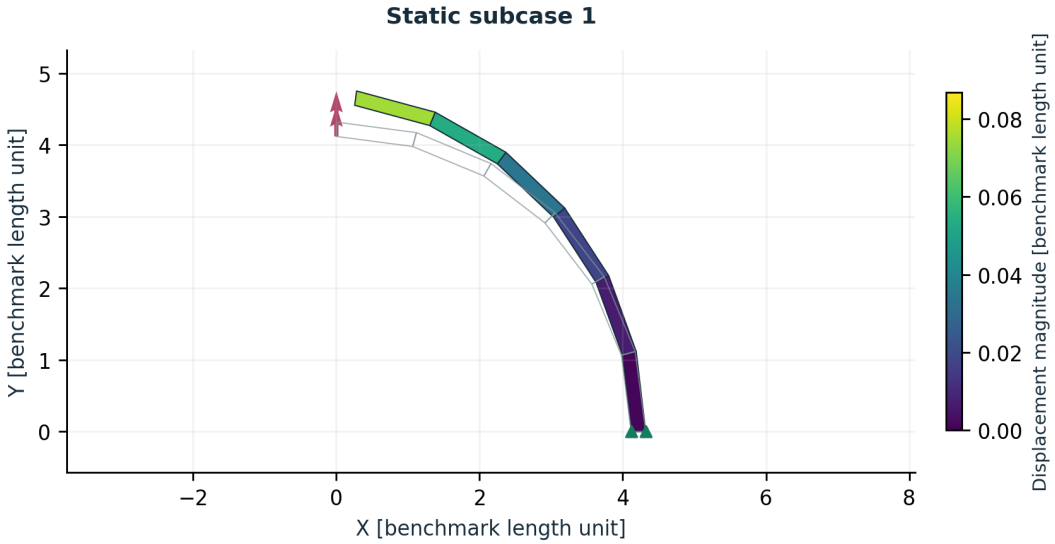
Curved cantilever: coarse mesh

MH_curved_beam_in_plane | SOL 101 | Published shell benchmark

90-degree annular strip; inner radius 4.12, outer radius 4.32, thickness 0.1; E=1e7, nu=0.25.

Geometry / mesh	14 nodes; 6 elements. 90-degree annular strip; inner radius 4.12, outer radius 4.32, thickness 0.1; E=1e7, nu=0.25.
Material / section	E=1e+07, nu=0.25; shell thickness=0.1. Consistent benchmark units; displacements in deck length unit, forces in deck force unit, stress in force/length^2
Boundary conditions	Both nodes at the start of the arc fixed in all 6 DOFs; unit resultant in global +Y at the free edge.
Applied load	Total Fy=1 at the free section, equally shared by its two edge nodes.



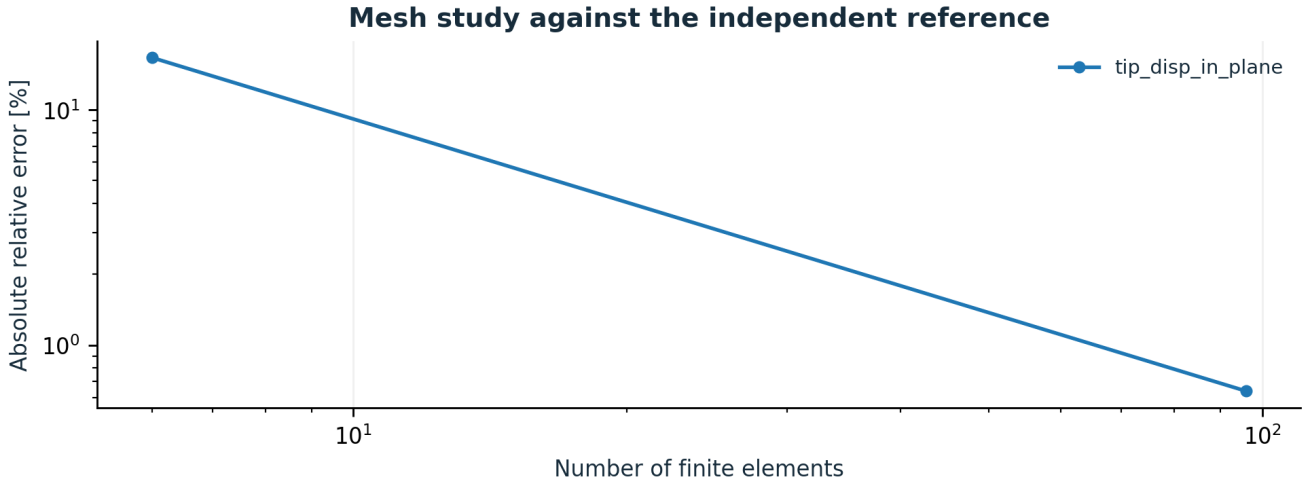
Shape amplified x5.97; grey = original.

Recorded JFEM result; deformation is amplified, mode amplitudes/signs arbitrary. Green triangles mark constrained nodes; magenta arrows mark available point loads.

Recorded numerical checks

Quantity [unit]	JFEM	Reference	Error	Assessment
tip disp in plane [benchmark length unit]	0.072787	0.08734	16.663%	FAIL (coarse)

Evidence, convergence and interpretation



The separately retained MH_curved_beam_refined companion tests the same published target. Coarse-mesh error is displayed and is not relabeled as validation success.

Scope: Linear elasticity and small-displacement equilibrium or linearized bifurcation/modal analysis; this is computational verification, not a material-test validation.

Reference and reproduction

MacNeal, R.H.; Harder, R.L. (1985). A proposed standard set of problems to test finite element accuracy. Finite Elements in Analysis and Design 1, 3-20. https://doi.org/10.1016/0168-874X(85)90003-4

Open the recorded .jfem file with Boxer: Project > Open JFEM results. For older eigenvalue files choose modal or buckling results when requested. The input deck and the recorded run receipt accompany the figures.

Run: 2026-10-19 | Solver revision: a70f997 | Full precision values: results.json / metrics.csv